



Smart mahogany remote

The Turn Touch smart remote may not have voice controls or a touch screen, but it is made out of solid mahogany <http://dgit.in/TurnTouch>



240 M year old fossil

Scientists have unearthed the fossil of a pregnant reptile that missed her due date by 240 million years. <http://dgit.in/240fossil>

Origins

this time who designed and constructed multiple iconic automatons of that era. In 1737, Vaucanson built the Flute player, The Tambourine Player and his most famous work, the Digesting Duck. This bird automaton had 400 moving parts in each wing (an astonishing number for that time) and it could actually flap its wings, consume water and grains and then defecate. This was achieved through a hidden compartment of fake 'digested food'. Such frauds were commonplace around the time as the onlookers mainly viewed automatons as means to entertain.



Da Vinci's design was way ahead of its time



YAY! THEY'RE MORE LIKE US NOW!

The 19th century saw multiple areas of progress that made advanced robotics (compared to the impractical automatons of the 18th century) a greater possibility. The automated Jacquard loom of 1801 and Babbage's Difference Engine in 1822 played key roles. In terms of theory, it was in 1847 that George Boole represented logic in a mathematical form understandable to machines, which is what we know as Boolean Algebra. Another significant development was the patenting of a remote controlled robot boat in 1898 by Nikola Tesla, which would form the basis of the wireless control that we see in space exploration robots even today.

It wasn't until the 20th century that the word 'Robot' was coined. In the 1921 play R.U.R (Rossum's Universal

Robots), writer Karel Capek used the word to denote automatons in the story. He credits the word to his brother, who derived it from the Czech word 'robota' that means servitude (kind of setting the intent right there). But a major, and perhaps the most significant, cultural phenomenon when it comes robotics was Asimov's science fiction writing in the 1940s. In fact, the word 'robotics' is said to have come into common parlance due to his work. Most importantly, he conceived the three laws of robotics:

- **First Law:** A robot may not injure a human being, or, through inaction, allow a human being to come to harm.
- **Second Law:** A robot must obey the orders given to it by human beings except where such orders would conflict with the First Law.
- **Third Law:** A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

He also added a zero law later:

- **Zeroth Law:** A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

In 1928, Japan's first humanoid robot, Gakutensoku, was designed and built by Makoto Nishimura, a biologist. It could perform some gestures like move its head and hand, and also change its expression. Another significant humanoid built during this time was Elektro, a robot built by the Westinghouse Electric Corporation somewhere between 1937 and 1938. This one could walk by voice command, speak about 700 words (using a record player), smoke cigarettes, blow balloons, and just like the Gakutensoku, move his head and arms.



HARDER, BETTER, FASTER, STRONGER

Around the 1950s, Alan Turing and his contemporaries did some remarkable work in computing and artificial intelligence. In 1950, Alan Turing devised the Turing test as a means to determine whether or not a machine has the ability to think for itself.

The first industrial robot, the Unimate (for 'universal automation')



Elektro with Sparko

was created by George Devol in 1954 and patented in 1961. It was actually used by General Motors in a plant in Trenton, New Jersey to lift hot pieces of die cast metal and stack them up. Unimation, the world's first robot manufacturing company – founded by Joseph F. Engelberger and George Devol in 1956 – is popularly regarded as the foundation of modern robotics.

The 1960s saw a lot of interest in robotics as an academic pursuit. Also, interest in Artificial Intelligence was rife around this time with AI research labs being opened at M.I.T., Stanford Research Institute (SRI), Stanford University, and the University of Edinburgh.

The world was also introduced to ELIZA and HAL. The first mobile robot that could comprehend and reason about its surroundings was also built this decade by SRI (now SRI International). Shakey, which is what the robot was named, navigated using a combination of multiple sensor inputs, including TV cameras, laser rangefinders, and "bump sensors".



Early UNIMATE model